Shahjalal University of Science and Technology, Sylhet
Department of Electrical & Electronic Engineering, SUST
Course Code: EEE 337 Course Title: Power System II
Term Test 1, Total Marks: 10, Time: 30 min

1. A double-circuit three-phase line is shown in Figure 1. The conductors a, a′; b, b′; c, c′ belong to the same phase, respectively. The radius of each conductor is 1.9 cm. Find the inductance of the double-circuit line in mH/km/phase. [5]

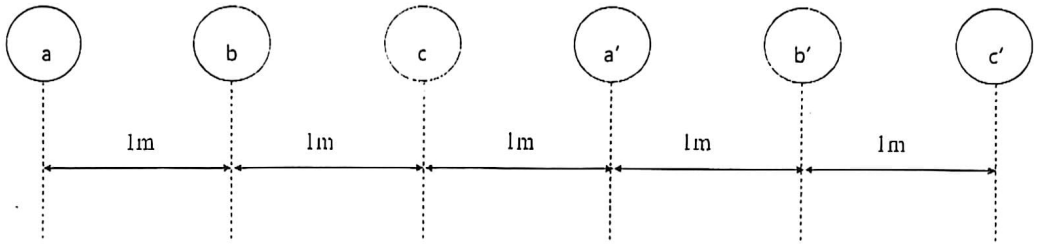


Figure: 1

2. Define transposition of a transmission line. Derive the expression for the capacitance of a three-phase line with unsymmetrical spacing. [5]